

Spatial Modelling of Satellite Data for the Development of a Wildfire Intensity Index in Italy

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1 Introduction and Objectives

Wildfires represent a major environmental and socio-economic challenge, affecting ecosystems, human activities and landscapes worldwide. In Europe, wildfire occurrence is particularly concentrated in Mediterranean regions, where the combination of climatic conditions, vegetation characteristics and human activities creates favourable conditions for fire ignition and spread. In the context of climate change, rising temperatures, prolonged droughts and longer fire seasons are expected to increase the frequency of conditions favourable to wildfire occurrence, although they do not directly cause fire ignitions. Because wildfire occurrence is governed by the interaction of multiple environmental and anthropogenic factors, characterising its spatial variability requires statistical models capable of integrating these drivers while accounting for residual spatial dependence. This thesis is situated within the context of long-term wildfire susceptibility modelling, aiming to describe the intrinsic spatial variability of wildfire occurrence rather than to provide short-term fire forecasts. Developed within a research internship at the Italian Institute for Environmental Protection and Research (ISPRA), this thesis aims to model the spatial intensity of wildfire occurrence across Italy using a Log-Gaussian Cox Process (LGCP). The proposed framework combines environmental, climatic, vegetation-related and anthropogenic covariates with a latent Gaussian spatial field, allowing the spatial log-intensity of wildfire occurrence to be estimated while quantifying the contribution of observed covariates and residual spatial variability to the observed wildfire pattern.

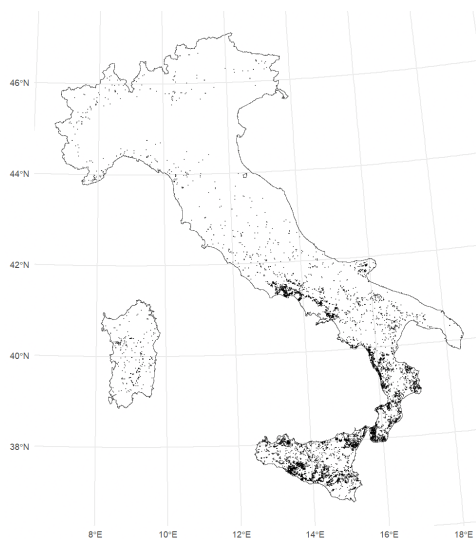


Figure 1: Final dataset of wildfire centroids

2 Data sources and Pre-processing

All datasets used in this thesis consist of standardized and harmonized Earth Observation products derived from the European Copernicus programme and national databases. This approach ensures spatial and methodological consistency across the Italian territory and provides a framework potentially scalable to larger geographical contexts.

2.1 BAD-ITE

The reconstruction of wildfire occurrence patterns is based on the Burnt Areas Database – Italian Terrestrial Ecosystems (BAD-ITE), developed by ISPRA from the satellite-based observations provided by the European Forest Fire Information System (EFFIS). The database was further enriched through integration with additional spatial information, including administrative boundaries and protected areas, to provide a more comprehensive characterization of wildfire events at the national scale. The analysis includes burned-area polygons mapped between 2018 and 2025 from Sentinel-2 observations, whose higher spatial resolution has improved the detection of smaller fire events and contributed to a more complete representation of burned areas. For the spatial point process analysis, the 9561 recorded wildfire events were converted from burned-area polygons into point locations through centroid extraction. The statistical domain was restricted to natural and semi-natural areas, consistently with the EFFIS definition of wildfires, by excluding artificial and densely urbanised areas through land-cover information derived from Copernicus products. Minor islands were also excluded to avoid artificial connections in the latent spatial field and improve the construction of the spatial integration scheme.

2.2 Environmental and Anthropogenic Covariates

The objective of the model is also to investigate the relationship between wildfire occurrence and a set of environmental and anthropogenic predictors. Accordingly, a collection of topographic, climatic, vegetation-related and human-pressure covariates was assembled to characterise the spatial heterogeneity of wildfire occurrence across Italy:

- **Landscape and spatial gradients:** elevation, latitude and distance from the coast were included to capture broad geographic variability and territorial gradients influencing environmental conditions and wildfire patterns.
- **Climatic conditions:** temperature, cumulative annual precipitation, average monthly precipitation, land surface temperature and solar radiation were considered to represent both long-term climatic regimes and recent (2018-2025) conditions affecting fuel moisture and fire-conducive environments.
- **Vegetation:** tree cover density was included as a proxy for vegetation structure and fuel availability.
- **Human-related factors:** population and human pressure index were included to represent spatial patterns of human presence and anthropogenic pressure.

All raster covariates were harmonized to a common spatial resolution of 1km and coordinate reference system and standardized before model fitting.

3 Methodological Framework: Spatial Point Processes and SPDE-INLA

Wildfire occurrences can be represented as a spatial point pattern, where the information of interest is contained in the geographical distribution of events across space. In this context, the objective is not to model fire severity, burned area conditional on occurrence, or potential impacts, but rather to characterise the spatial structure underlying wildfire occurrence. Each observed event represents a

point location within the study domain and the main quantity of interest is the spatial intensity function ($\lambda(s)$), which describes the expected number of wildfire events per unit area over the observation period. Estimating how this intensity varies across space allows the identification of areas characterised by higher or lower underlying occurrence intensity.

A Log-Gaussian Cox Process (LGCP) was selected because it provides a suitable framework for modelling spatial point patterns generated by complex environmental processes. Wildfire events are not spatially independent, as neighbouring locations often share similar climatic conditions, vegetation characteristics, landscape properties and human-related factors. The LGCP accounts for this behaviour by modelling the intensity as a function of both observed covariates and an underlying latent Gaussian spatial field, separating the contribution of measured predictors from residual spatial dependence associated with unobserved processes.

The general formulation of the model is:

$$\log(\lambda(s)) = \beta_0 + \sum_{k=1}^p \beta_k x_k(s) + \sum_{j=1}^q f_j(z_j(s)) + w(s)$$

where:

- β_0 is the intercept;
- $x_k(s)$ are covariates included with linear effects;
- β_k are the corresponding coefficients;
- $f_j(\cdot)$ are any non-linear or smooth effects;
- $z_j(s)$ are covariates modelled non-linearly;
- $w(s)$ is a zero-mean Gaussian random field capturing residual spatial dependence.

This formulation allows the spatial pattern of wildfire occurrence to be interpreted as the result of both observed environmental and anthropogenic factors and unobserved spatially structured processes.

3.1 The SPDE-INLA Computational Approach

The estimation of LGCP models over large spatial domains is computationally challenging because Bayesian inference requires handling a high-dimensional latent spatial field. To overcome this limitation, the model was implemented using the SPDE-INLA framework. The SPDE approach provides a computationally efficient representation of the continuous Gaussian random field by approximating it through a discretised Gaussian Markov Random Field (GMRF). This is achieved by defining an irregular triangular mesh over the study region, whose resolution can adapt to the spatial complexity of the domain and the distribution of observations. Compared with regular grid approximations, irregular meshes provide greater flexibility in representing complex geographical boundaries and allow computational resources to be concentrated where higher spatial resolution is required.

The key advantage of the SPDE formulation is that the resulting GMRF is characterised by a sparse precision matrix due to the local dependence structure between neighbouring mesh nodes. This sparse representation substantially reduces memory requirements and computational costs, making spatial models based on continuous domains feasible at national scales.

Within this framework, Bayesian inference was performed using Integrated Nested Laplace Approximation (INLA). Markov Chain Monte Carlo (MCMC) represents the most widely used and flexible approach for Bayesian inference, as it can be applied to a broad range of models by approximating posterior distributions through stochastic sampling. However, in spatial hierarchical models such as LGCPs, the presence of a high-dimensional latent Gaussian field can make MCMC computationally demanding, as a large number of latent variables must be estimated and convergence may require substantial computational time.

INLA provides an efficient alternative specifically designed for latent Gaussian models. It is a deterministic algorithm based on nested Laplace approximations that directly estimates marginal posterior distributions of model parameters and hyperparameters, rather than approximating the full joint posterior distribution through sampling. This approach substantially reduces computational costs while maintaining accurate Bayesian inference, making it particularly suitable for large-scale spatial models involving complex latent structures.

The combination of LGCP, SPDE and INLA therefore provides a suitable framework for wildfire occurrence modelling, as it preserves the continuous spatial nature of the point process, accounts for residual spatial dependence and enables efficient Bayesian estimation of both covariate effects and latent spatial variability at the scale of the Italian territory.

3.2 Model selection

Model selection was performed to identify the LGCP specification providing the best compromise between goodness of fit, predictive performance and model complexity. Before model comparison, an initial screening of the candidate predictors was performed to reduce redundancy and limit the effects of multicollinearity. In particular, highly correlated variables representing similar processes were not included simultaneously within the same candidate model.

A set of competing LGCP formulations was then fitted by systematically exploring different combinations of non-redundant predictors. Model comparison was primarily based on the Watanabe–Akaike Information Criterion (WAIC), with the Deviance Information Criterion (DIC) used as an additional diagnostic measure. The stability and interpretability of the estimated covariate effects were also considered through the analysis of posterior credible intervals.

Since information criteria alone do not always provide a unique optimal solution, especially for complex latent Gaussian models including spatial random effects and non-linear components, the final model was selected by jointly considering statistical performance, ecological consistency and model parsimony. After identifying the most competitive covariate combinations, alternative functional forms were evaluated to determine whether non-linear effects provided a substantial improvement over linear specifications.

4 Main Results and Effects of Covariates

Model comparison based on WAIC and DIC indicated that allowing flexible non-linear effects substantially improved model performance with respect to purely linear specifications. These results suggest that the relationships between wildfire occurrence intensity and several environmental and anthropogenic drivers cannot be adequately represented by simple monotonic trends. Its structure is given by:

$$\begin{aligned} \log \lambda(s) = & \underbrace{\beta_0 + \beta_1 \text{Temp}(s)}_{\text{Fixed effects}} \\ & + \underbrace{f_{\text{elev}}(\text{Elev}(s)) + f_{\text{coast}}(\text{Coast}(s)) + f_{\text{tree}}(\text{Tree}(s)) + f_{\text{popul}}(\text{Popul}(s))}_{\text{Non-linear effects (RW2)}} \\ & + \underbrace{w(s)}_{\text{Latent Gaussian spatial field}}, \quad s \in D \end{aligned} \quad (1)$$

The estimated effects of the selected covariates are summarised in Table 1 .

Table 1: Summary of the covariates included in the final LGCP model.

Covariate	Effect	Main interpretation
Temperature	Linear	Positive association with wildfire occurrence intensity. Areas characterised by persistently warmer climatic conditions show higher estimated log-intensity after accounting for the remaining covariates and the latent spatial field.
Elevation	RW2	Strong non-linear relationship. Wildfire occurrence intensity is highest at intermediate elevations (approximately 600–750 m a.s.l.), while both lowlands and high mountain environments exhibit lower estimated intensity.
Tree cover density	RW2	Non-monotonic effect. Intermediate tree cover (approximately 20–40%) is associated with the highest log-intensity, whereas densely forested areas show progressively lower estimated intensity.
Distance from coast	RW2	Complex non-linear relationship. The effect varies along the coastal–inland gradient, suggesting that this variable acts as a proxy for multiple environmental and anthropogenic processes rather than having a direct causal interpretation.
Population	RW2	Strongly non-linear effect. Wildfire occurrence intensity is highest at intermediate levels of surrounding human presence and decreases in areas characterised by very high value of population.

The final model highlights the importance of allowing flexible covariate effects. Temperature is the only predictor adequately described by a linear relationship, whereas elevation, tree cover density, distance from the coast and population exhibit marked departures from linearity. These results indicate that wildfire occurrence intensity is driven by complex interactions between environmental gradients and human influence, which cannot be captured by simple linear effects. The latent spatial field is characterised by a posterior mean spatial range (ρ) of approximately 47 km, indicating that residual spatial dependence mainly operates at a sub-regional scale. Moreover, the posterior distribution is concentrated on substantially shorter ranges than suggested by the prior specification, showing that the data support a relatively local dependence structure. The posterior marginal standard deviation ($\sigma \approx 2.5$) indicates considerable residual spatial variability after accounting for the observed covariates, suggesting that additional unmeasured environmental or anthropogenic factors still contribute to the observed wildfire pattern.

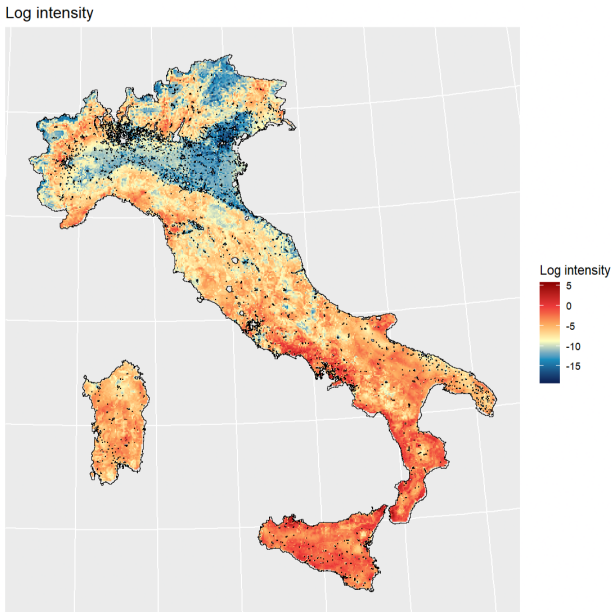


Figure 2 shows the posterior mean of the estimated spatial log-intensity, obtained by combining the effects of the selected covariates and the latent Gaussian spatial field. The estimated surface represents the logarithm of the expected wildfire occurrence intensity per unit area over the study period and therefore provides a relative description of the spatial distribution of wildfire occurrence. The estimated log-intensity reveals a pronounced spatial heterogeneity across Italy, with a clear north–south gradient. Higher values are concentrated in southern and insular regions, particularly Sicily and Calabria, while lower values are generally observed across the Alpine arc and the Po Valley. This large-scale pattern is not spatially uniform, however, as localized areas of relatively high log-intensity also emerge in parts of central Italy and along the Alpine foothills. Overall, the estimated surface is consistent with the effects identified for the environmental covariates, indicating that warmer climatic conditions and intermediate elevations are associated with higher wildfire occurrence intensity.

Figure 2: Estimated log intensity of wildfires occurrence

Log intensity latent gaussian field

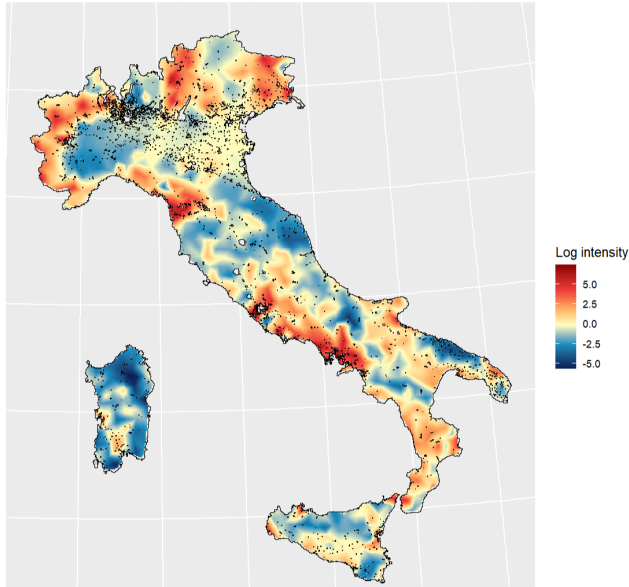


Figure 3: Estimated log intensity of latent gaussian random field

5 Final considerations

The proposed LGCP framework provides a data-driven approach for characterising the long-term spatial intensity of wildfire occurrence across Italy. By modelling wildfire events as a spatial point pattern, that is, as the exact geographical locations of wildfire occurrences rather than as aggregated counts, the analysis preserves the spatial information contained in the observed events and allows the contribution of environmental and anthropogenic drivers to be quantified while accounting for residual spatial dependence. The use of harmonised Earth Observation products represents an additional strength of the approach, as it ensures consistency across the study area and supports the potential application of the same framework to other European regions where comparable datasets are available.

Beyond the production of spatial intensity maps, one of the main advantages of this methodology is its ability to investigate how different drivers are associated with wildfire occurrence without imposing predefined relationships. Unlike approaches in which risk classes or weights are assigned a priori to environmental variables, the statistical framework allows the observed wildfire pattern to inform the estimated relationships between covariates and occurrence intensity. This provides a flexible tool for exploring complex ecological relationships, including non-linear responses that may otherwise remain undetected.

The analysis highlighted that the broad spatial gradient of wildfire occurrence across Italy is largely explained by the included environmental and anthropogenic variables, while the latent spatial field captures additional local-scale variability. In particular, the residual patterns observed in some northern regions represent an interesting direction for future research. These deviations may reflect unmeasured environmental processes, limitations in the spatial and temporal resolution of the available climatic predictors, or other local factors not represented in the current model. Further investigation, potentially incorporating higher-resolution climatic information and additional explanatory variables, would be required to identify the mechanisms underlying these patterns.

Several limitations should be acknowledged. First, the model describes long-term spatial variation in wildfire occurrence intensity and does not account for the temporal dynamics associated with individual fire events. Some variables, such as precipitation, may provide additional information when considered dynamically, for example through short-term anomalies preceding fire occurrence, rather than as long-term spatial descriptors. A natural extension of this work would therefore involve the development of a spatio-temporal LGCP framework capable of capturing seasonal variability, climate-driven changes and temporal variations in human activity.

Second, wildfire locations were represented through burned-area centroids rather than ignition points. Although appropriate for modelling spatial occurrence patterns at the national scale, this approximation may influence the relationship with anthropogenic variables, particularly those describing accessibility and human presence. The availability of precise ignition locations would allow a more detailed assessment of the role of roads, settlements and infrastructures in wildfire occurrence. Similarly, extending the response variable to include burned area or fire severity would allow the analysis to move from occurrence intensity towards a more comprehensive characterisation of wildfire impacts.

Finally, future developments should include an external validation phase to assess the predictive performance and transferability of the model. Such validation would provide additional evidence on the ability of the framework to generalise beyond the observed period and study domain.

Overall, the spatial intensity maps produced in this study provide a rigorous framework for analysing the spatial heterogeneity of wildfire occurrence in Italy. By identifying environmental conditions associated with increased occurrence intensity and highlighting areas with unexplained residual patterns, the results can contribute to improved understanding of wildfire dynamics and support future applications in land management, prevention planning and allocation of fire suppression resources.